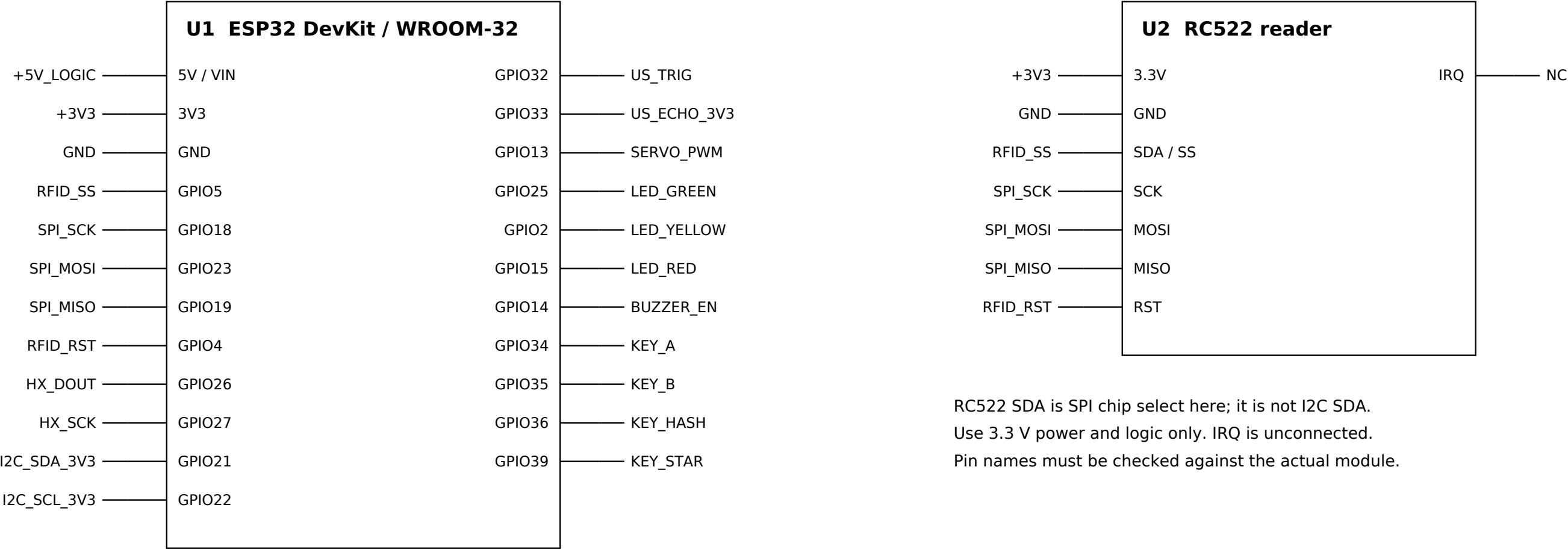
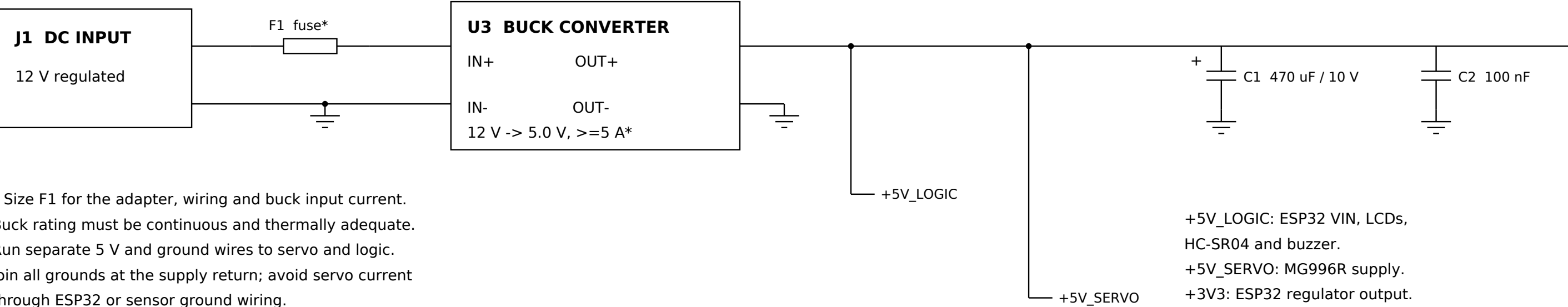


SMART BCS

Controller, RFID and power distribution

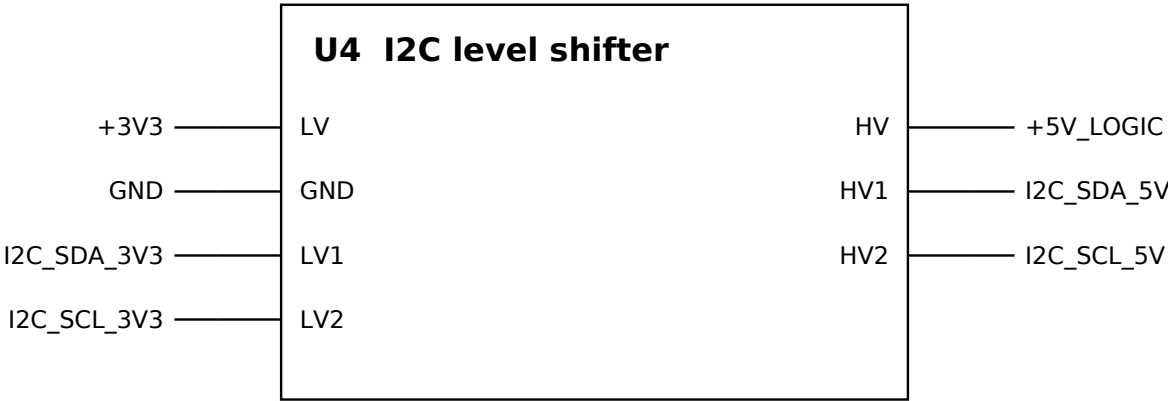


POWER INPUT AND SEPARATE LOAD BRANCHES

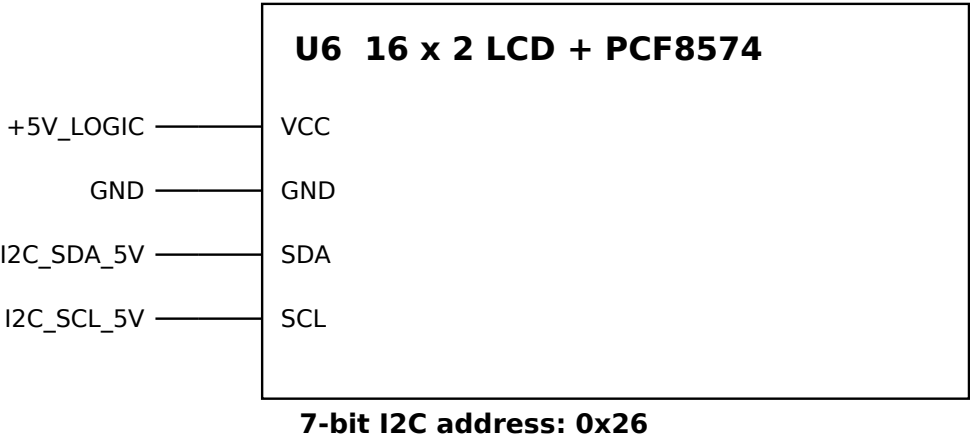
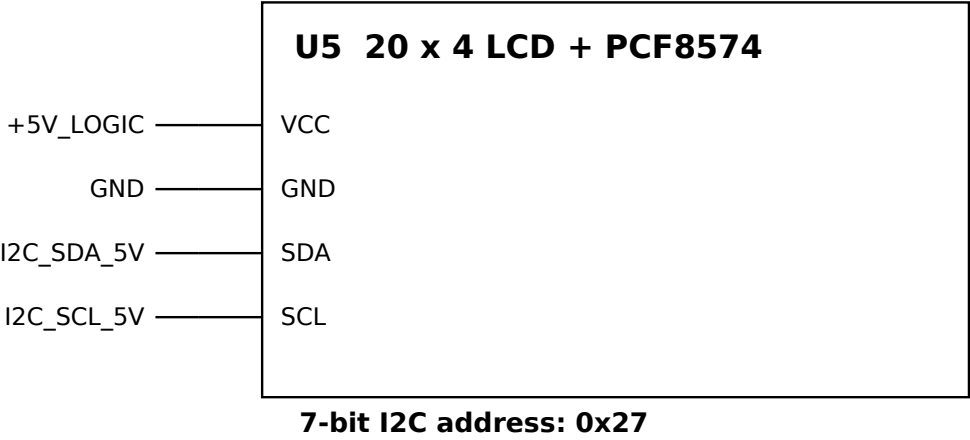


Matching net names are electrically connected across all three sheets. Dots indicate junctions; crossings without dots do not connect.
Do not feed external 5 V and USB 5 V simultaneously unless your DevKit has verified power isolation.

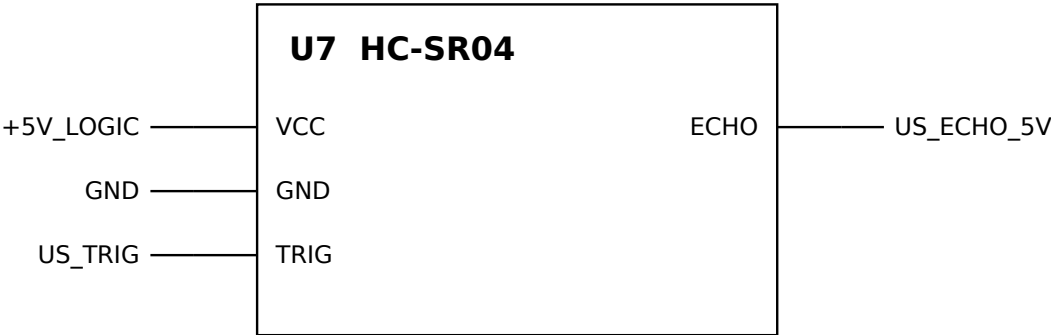
I2C BUS - 3.3 V / 5 V LEVEL TRANSLATION



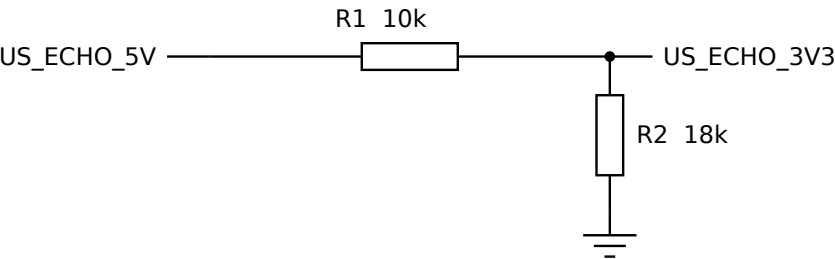
U4: two-channel bidirectional, open-drain I2C translator, e.g. BSS138 module with pull-ups on both voltage sides. Use about 4.7k effective pull-up per line per side; account for resistors already fitted to the LCDs and translator. Verify addresses with an I2C scan; use a 100 kHz bus. Do not connect 5 V LCD pull-ups directly to ESP32 GPIO.



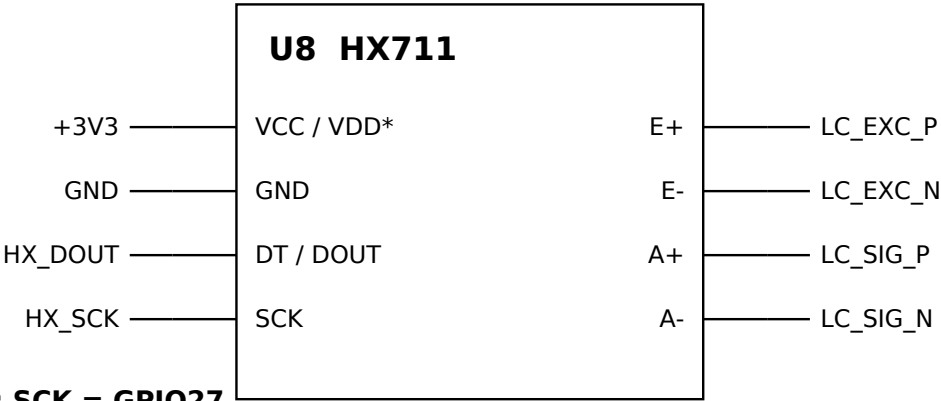
ULTRASONIC SENSOR AND ECHO DIVIDER



Echo divider output: about 3.21 V for a 5 V echo. Bin-level monitoring: TRIG = GPIO32; ECHO = GPIO33.

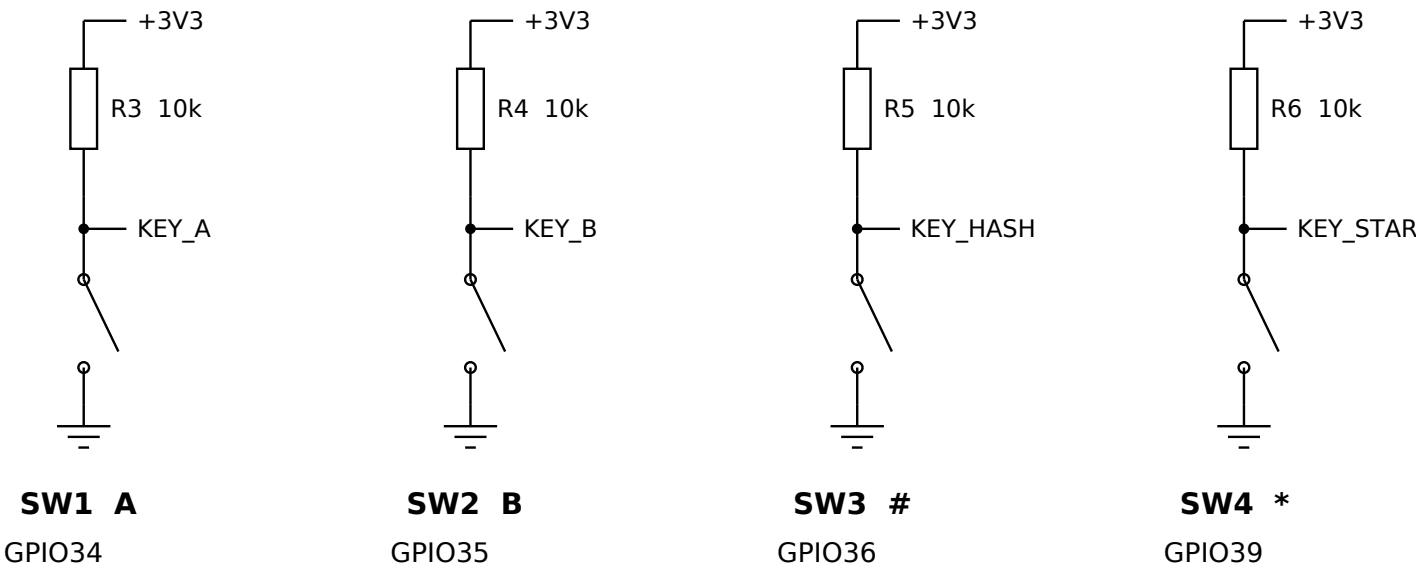


LOAD CELL AND HX711



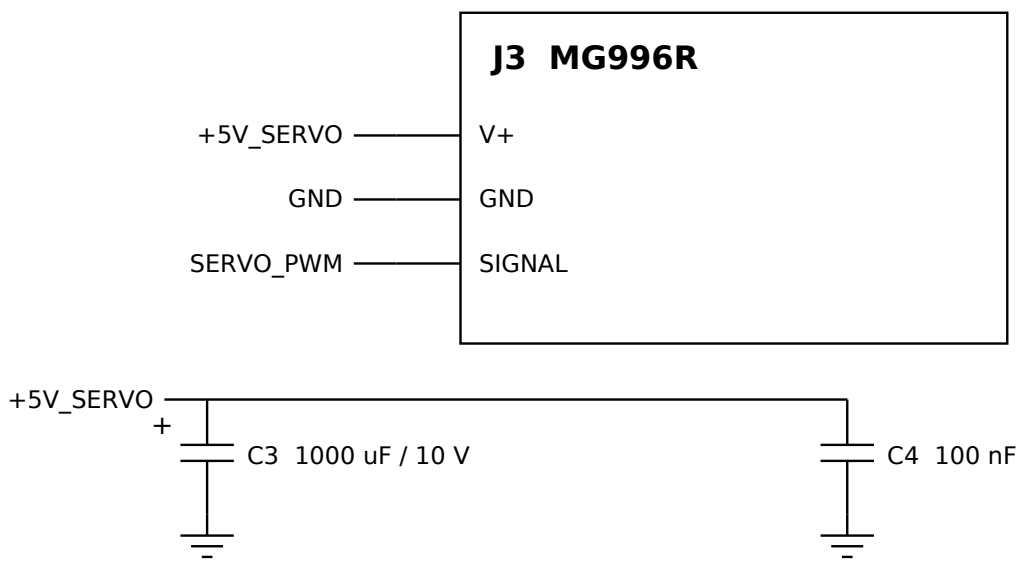
DOUT = GPIO26; SCK = GPIO27.
J2 Four-wire load cell: E+ -> LC_EXC_P; E- -> LC_EXC_N; A+ -> LC_SIG_P; A- -> LC_SIG_N. B+ and B- unused.
* For common single-supply HX711 boards, use 3.3 V. For separate VCC/VDD boards, use 3.3 V for both here. Verify wire functions from the load-cell datasheet; recalibrate.

FOUR INDEPENDENT NORMALLY OPEN SWITCHES



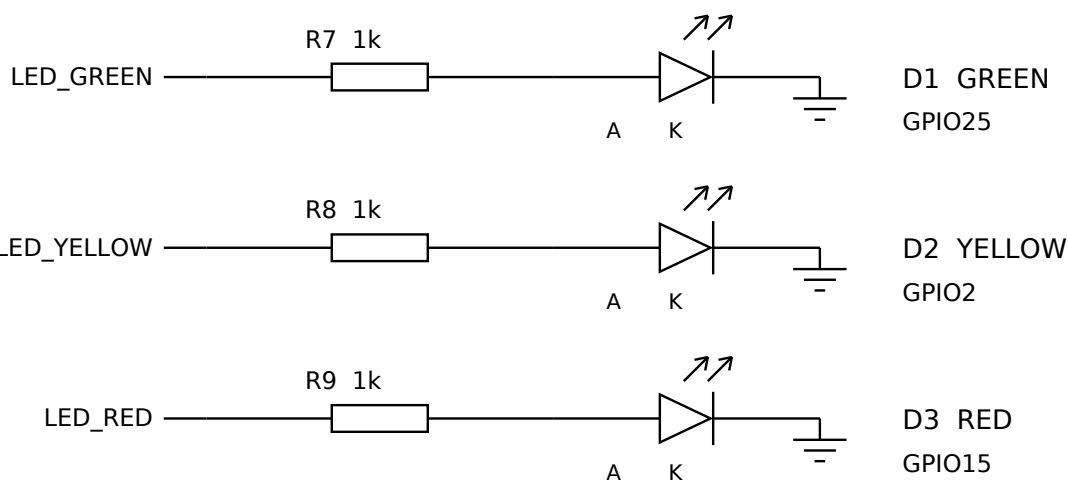
Pressed = LOW. Each key closes only its own GPIO to ground.
All four keys use input-only GPIOs; fit the 10k pull-ups shown.
Four-pin pushbuttons have internally joined pairs; verify with a meter.

SERVO CONNECTOR - GPIO13



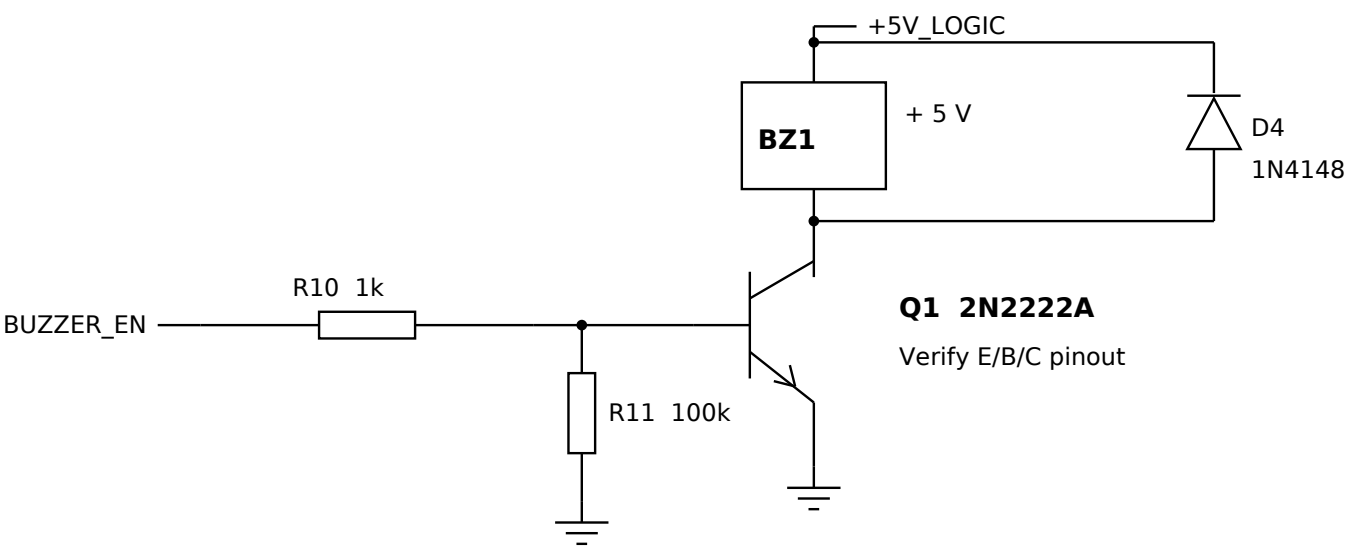
Fit capacitors near servo connector. Allow ≥ 2 A supply capacity, with margin for actual stall current. Confirm that this servo accepts a 3.3 V control signal.

LED INDICATORS - ACTIVE HIGH



GPIO2, GPIO5 and GPIO15 are ESP32 strapping pins.
Keep external circuitry from overriding boot strap levels.
LEDs use series resistance; no direct LED-to-GPIO wiring.

BUZZER DRIVER - GPIO14 / ACTIVE 5 V



Driver assumes buzzer current ≤ 30 mA; D4 cathode connects to +5 V.

FINAL PIN MAP: A = 34; B = 35; # = 36; * = 39. Servo = 13; buzzer = 14. GPIO assignments match the user-confirmed working mapping.
Sources: Espressif ESP32 datasheet; NXP PCF8574/PCF8574A datasheet; SparkFun HX711 hookup guide. Hardware has not been bench-tested.